Project Report

Smart Medicine

Presented By

Teerapat Kotanart 6604062636348

Present to

Associate Professor Kobkiat Saraubon, Ph.D.

This report is part of the Internet of Things subject.

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King Mongkut's University of Technology North Bangkok

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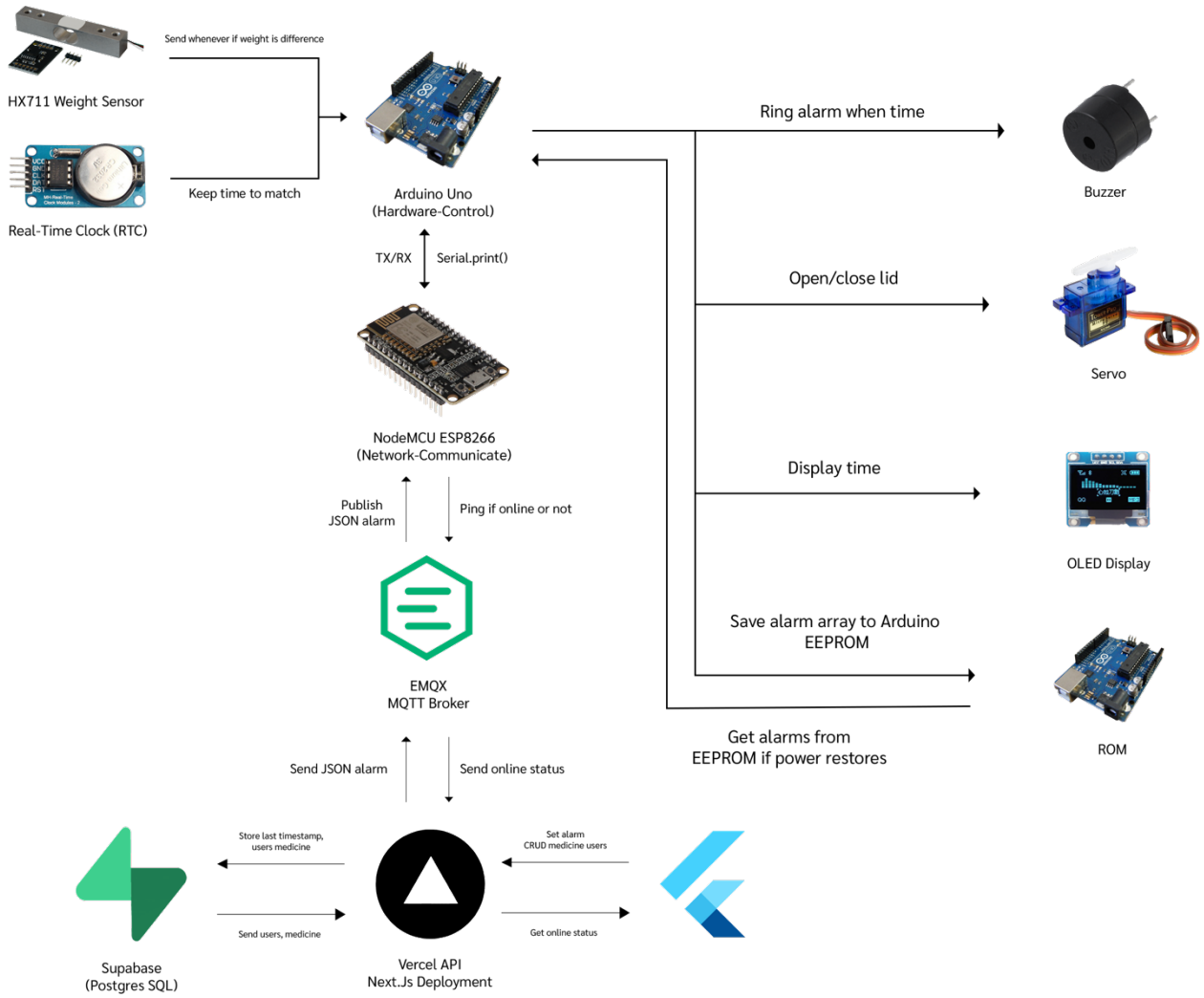
Section 1: Overview

This smart medicine device is designed for patient and can manage their daily medication schedules. If who someone have a patient and them forget to take their medicine this device can manage and arrange the drugs for patient with mobile application. This device can help your patient eat drugs in time and set the time for each drug.

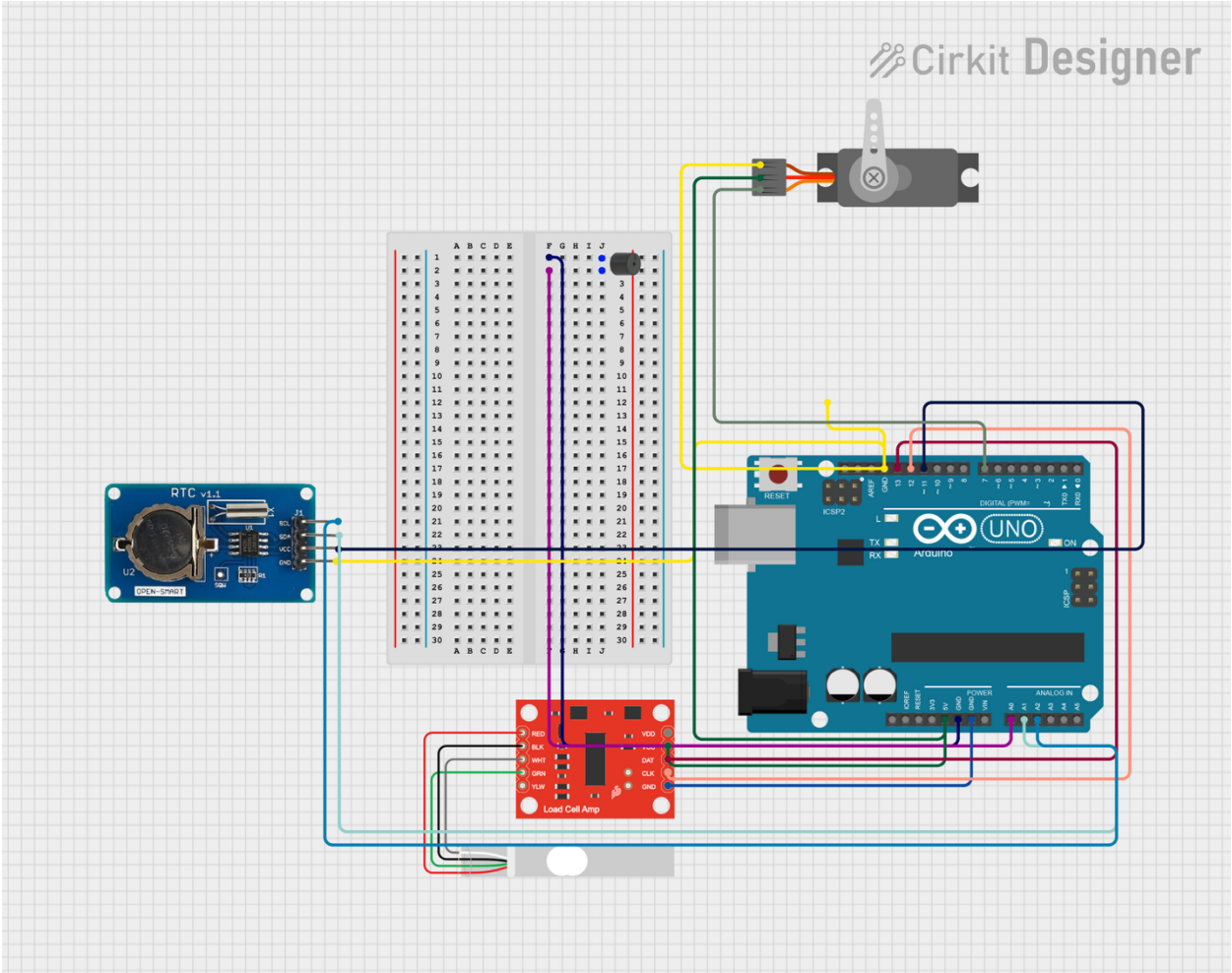
The system in this device using esp8266, Arduino Uno, sensors, and smart medicine box device connected through MQTT, can add, update, or remove the drug through mobile application, additionally this app can follow the schedules drug of patients what have how many drug patients must to eat.

Conclude, the smart medicine improves medication adherence, reduce the risk of missed doses, and provide a convenient monitoring tool for caregivers.

Section 2: System Architecture



Section 3: Diagram



Section 4: Data Structure

Collect data, this system we create data structure alarm for collect name drug, time alert, status alert

```
struct Alarm {
  char name[NAME_LEN];
  byte h;
  byte m;
  bool triggered;
};
```

```
void callback(char* topic, byte* payload, unsigned int length)
```

this is communication logic and can receive command from MQTT, convert command and send to system

Receive command from IoT Platform e.g.,

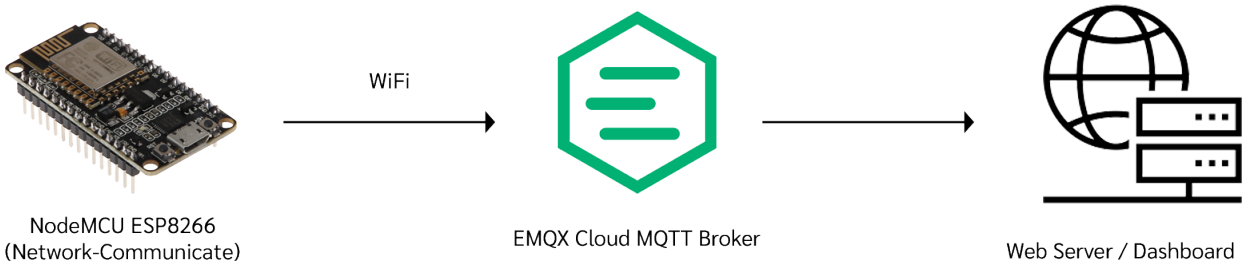
```
medicine/set_alarm
medicine/refill
medicine/tare
```

And convert was command for Arduino process

Section 5: IoT Platform / Service

In this project we use EMQX Cloud platform it was a Cloud MQTT Broker for system IoT it can use to send-return data between IoT device e.g., ESP8266, Arduino and application system via internet and user protocol from IoT device

For example, how to send data between Device with IoT platform



Send data (Publish)

Device Smart Medicine Box will send status to MQTT Topic

```
medicine_box/status
```

For example, data

```
{
        "weight": "-4",
        "time": "2026-03-08 12:00:01"
}
```

And device can receive command from Cloud system via Topic e.g.,

```
medicine/set_alarm
medicine/clear_alarm
medicine/refill
medicine/tare
```

How to setup

1. Create Deployment on EMQX Cloud
2. Create new project
3. Create Deployment (MQTT Broker)
4. And then if created system will show connect data e.g.,

```
Broker Address: l2901b8a.ala.asia-southeast1.emqxsl.com
Port: 8883
Protocol: MQTT over TLS
```

IoT device e.g., Arduino + ESP8266 will connect via WiFi by MQTT Client Library e.g., PubSubClient

```
client.setServer("l2901b8a.ala.asia-southeast1.emqxsl.com", 8883);
client.connect("medicine_box","username","password");
```

When you connected device it can

1. Publish data to MQTT Topic
2. Suvscribe for receive command from Server

Section 6: Example code sensors

Buzzer

```
const int infrared = D0;
const int led = D5;
const int buzzer = D1;

void setup() {
  pinMode(infrared, INPUT);
  pinMode(led, OUTPUT);
  pinMode(buzzer, OUTPUT);
}

void loop() {
  int motion = digitalRead(infrared);

  if ( motion == LOW ) {
    digitalWrite(led, 1);
    // open sound
    tone(buzzer,3000);
  } else {
    digitalWrite(led, 0);
    // sound off
    noTone(buzzer);
  }

  delay(50);
}
```

Buzzer can make sound for alert patient must eat drug when in time

```
      tone(PIN_BUZZER, 400, 200);
```

And if you want to set smooth sound you can do like this

Section 6: Example code sensors

Servo

```
#include <Servo.h>

Servo myServo; // Create servo object
const int servoPin = 9; // Pin connected to servo signal

void setup() {
  myServo.attach(servoPin); // Attaches the servo on pin 9
  myServo.write(0);        // Set initial position to closed (0 degrees)
  delay(1000);             // Wait for it to reach position
}

void loop() {
  // --- OPEN DOOR ---
  myServo.write(120);       // Rotate to 90 degrees to open
  delay(1000);             // Keep door open for 5 seconds

  // --- CLOSE DOOR ---
  myServo.write(0);        // Rotate back to 0 degrees
  delay(1000);             // Wait 5 seconds before opening again
}
```

Servo 180 can rotate and fix degrees for open the box and close the box

Section 6: Example code sensors

HX711

```
#include "HX711.h"

#define DOUT 2
#define CLK 3

HX711 scale;

float calibration_factor = -7050;

void setup() {
  Serial.begin(9600);

  scale.begin(DOUT, CLK);
  scale.set_scale(calibration_factor);
  scale.tare();

  Serial.println("HX711 Ready...");
}

void loop() {

  float weight = scale.get_units(10);

  Serial.print("Weight: ");
  Serial.print(weight);
  Serial.println(" g");

  delay(200);
}
```

HX711 can check weight of drugs for check patient eat drug yet, this code is example for check weight of something you put above of HX711 device

Section 7: Appendix

Appendix A: MQTT Callback Function

```
void callback(char* topic, byte* payload, unsigned int length) {

  String msg = "";

  for (int i = 0; i < length; i++)
    msg += (char)payload[i];

  String t = String(topic);

  if (t == "medicine/set_alarm") {
    enqueueCommand("CMD:ADD_MED:" + msg);
  }

  else if (t == "medicine/clear_alarm") {
    enqueueCommand("CMD:CLEAR_MED");
  }

  else if (t == "medicine/refill") {
    enqueueCommand("CMD:REFILL");
  }

  else if (t == "medicine/tare") {
    enqueueCommand("CMD:TARE");
  }
}
```

This function handles MQTT messages received from the IoT platform. It reads the topic and payload, then converts them into internal commands for the medicine box system.

Appendix B: MQTT Topic Structure

Topic	Function
medicine/set_alarm	Set medicine alarm
medicine/clear_alarm	Clear all alarms
medicine/refill	Refill medicine
medicine/tare	Reset load cell weight

These topics are used for communication between the IoT platform and the smart medicine box device.